No of Pages : 4 **Course Code : 23MX15**

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, DECEMBER 2023

MCA Semester : 1

23MX15 WEB TECHNOLOGIES

Time : 3 Hours **Maximum Marks : 100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Subdivisions (a) carries 3 marks each, subdivision (b) carries 10 marks each and subdivision (c) carries 12 marks each.
3. Course Outcome : Table

Qn.1	CO1	Qn.2	CO2	Qn.3	CO3	Qn.4	CO4

1. a) Determine the Network mask, host mask, network id and host id of all the public IP addresses only among the given IP addresses.
   i) 127.0.0.21/8 ii) 145.68.1.23 / 14 iii) 192.32.40.21 / 21

   b) i) Why is multiplexing/demultiplexing needed in Transport layer and Network layer? (4)

   ii) Compare and contrast the different methods involved in making the HTTP request. Also explain what do the code 2xx and 5xx generated by HTTP response mean? (6)

   c) i) What are web browsers? What are the different components of any web browsers? Describe the working of different phases of a rendering engine.

   (OR)

   ii) Why do we need domain names even though there are 128 bit IPv6 addresses? Explain the process of obtaining equivalent IP address for the given domain name www.amazon.com assuming that it is not cached at the local DNS server. Describe all the steps taken by the domain name system in resolving the IP address for www.amazon.com .

2. a) State the significance of metadata in a HTML file. Where is it provided in a HTML file?

   b) Describe with an example the different layout models used in CSS to make your web page presented in different format.

   c) Write the proper HTML and external CSS code to design the following form. Upon clicking the signup button and all field entries are correct an message " SignUp successful "should be displayed.

3. a) What is hoisting in JavaScript? What will be the output/error(s) of the following code snippet? Justify your answer.

```
var a = 5;
( function() {
console.log(a);
var a = 10;  } ) ( ) ;
```

b) i) Analyze the following code, trace the output/ error and explain your answer. (4)

```
<!DOCTYPE html>
<html>   <body>
<div  onmousedown="mDown(this)" onmouseup="mUp(this)"
      style="background-color:green ; width:90px ; height:20px ; padding:40px ;
            color:blue">   Click Me  </div>
<script>
function mDown(obj) {
 obj.style.backgroundColor = "yellow";
 obj.innerHTML = "Release Me";  }
 function mUp(obj) {
 obj.style.backgroundColor="Orange";
 obj.innerHTML="Thank You";       }
</script>   </body>  </html>
```

ii) Write a dynamic HTML program to generate the following table to be rendered when user clicks the button "Start" on the screen. Note: The table should be generated at run time not by changing its attribute from hidden to display. (6)

Date	Time	Status
5.1.2023	5.15PM	"Holiday mood"

c) Write the proper class based JavaScript program to simulate the following class hierarchy. This inheritance model is to be used for analyzing the student marks earned on the daily quiz conducted in a class. myScores class has dailyScore array to store the scores earned daily. Write the proper JavaScript code

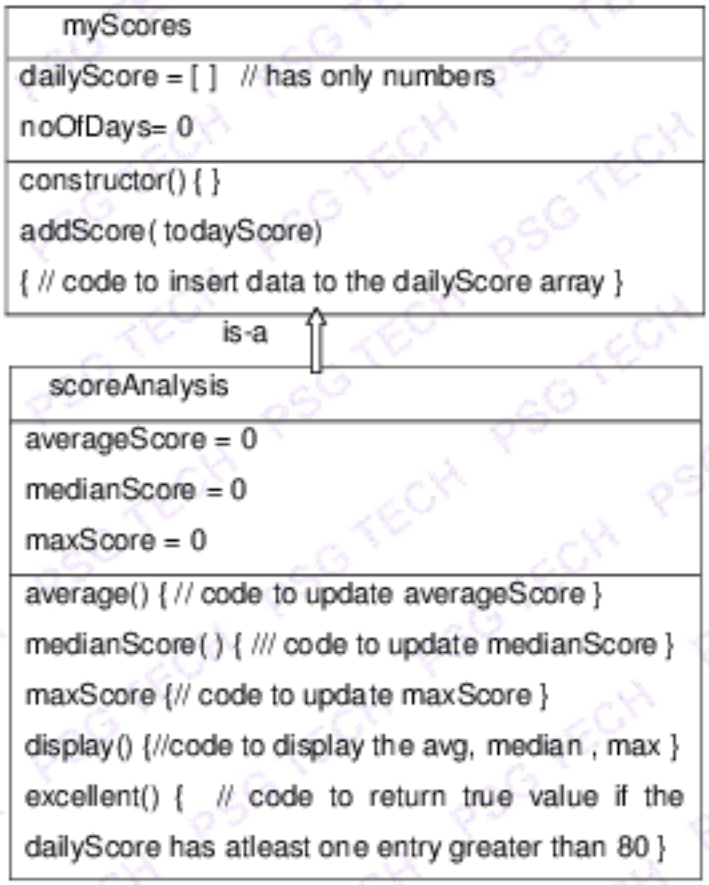
i) to create two objects of scoreAnalysis for stud1 and stud2.

ii) to add the scores of stud1 and stud2 for five quizzes.

iii) Display the maximum score of both the students.

iv) Check whether they have any excellent score or not.

v) Display the person with highest average score.



4. a) Analyze the following code and write the output/error generated. **Note:** Line numbers are shown for your easy reference. (3)

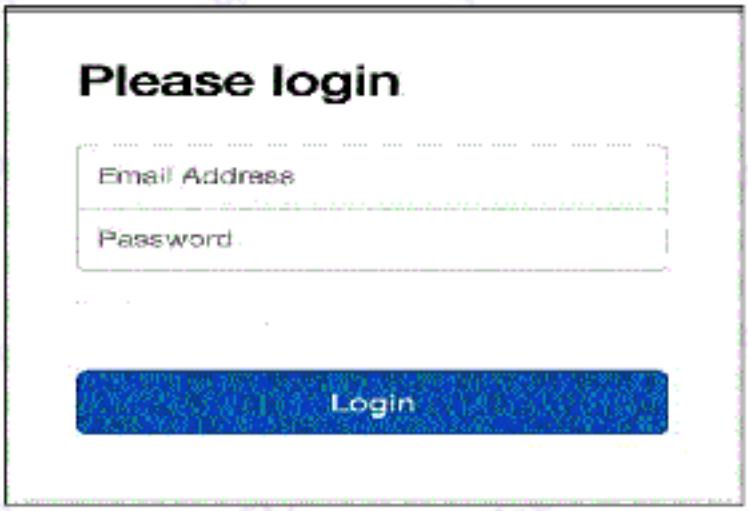
```
var count=0;
function job(myEvent) {  count++;
        if (count%2) myEvent=false;
return new Promise(function(resolve, reject) {
    if (myEvent) { resolve("success"); }
    else  reject();                    });        }
let promise = job(true);
promise .then(function(soln) {  console.log(soln);        } )
.then(function() {   console.log("Success");     } )
.catch(function() {      console.log("Error");     } )
.then(function() {     console.log("Success");      } );
```

b) Describe in detail all the features of NodeJS and the significance of its worker threads. Also justify the statement "Even though the NodeJS is single threaded environment it is highly scalable" with proper explanation and diagram. (10)

c) Write a proper MongoDB based NodeJS application code to develop the following login page for your project. Do proper validation for email fields. On login being success display the message

"Welcome <user email> . You have Successfully logged in.".

Note: The user login email and password is to be verified from the "user" collection in the database.



Write all the programming code needed to develop this page, code for validating the form and displaying the success message. Explain the flow of your code. (12)

/END/

FD/JU